multi-cb
PRINTED CIRCUIT BOARDS
1-48 layers · From 1WD express · 30 years

Materials Default Values – FR4

Poole, 15.10.2025

In the following, we have listed the established default values for FR4 materials used at Multi Circuit Boards Ltd.

The quality of your PCBs is assured through the use of technical comparable materials (dependent on manufacturing inventory, global market). A specific material can be defined on request; this may lead to a longer production time and / or a price adjustment.

		FR4 Standard	FR4 High-TG	FR4 Halogen-free	FR4 Track **resistant**	Polyimid
TG	°C	≥ 135	≥ 170	≥ 140	≥ 135	≥ 230
CTE-z	ppm/°C	60	45	45	50	20
Permittivity εr	at 1MHz	4,6	4,8	4,7	4,7	3,4
Electrical strength	KV/mm	45	55	50	45	125
Surface resistivity	MΩ	10^6	10^7	10^7	10^7	10^6
Tracking resistance	V	175 - 249*	175 - 249*	175 - 249*	400 - 599**	-
CTI	PLC	3	3	3	1	-
Td value	°C	301	345	360	310	***
Peel strenth	N/mm	1,1 - 1,6	1,2	1,4	1,6	1,6
Flammability class	UL-94	V0	V0	V0	V0	V0

Standard		IPC-4101	IPC-4101	IPC-4101	IPC-4101	IPC-4204
Depending on availability and / or customer request	e.g.	Isola Duraver Shengyi S1141 KingBoard 6160 Panasonic R-1766M	Isola IS410 Shengyi S1000-2 Iteq IT180-A Panasonic R-1755V	Isola Duraver 156 Shengyi S1155 KingBoard 6162 Panasonic R-1566V	Isola Duraver KF Shengyi S1600	DuPont Pyralux Shengyi SF302 Panasonic R-F775

* ISOLA recommends the value of 200 for calculations.
** on request PLC 0 ≥ 600V
*** Recommended operating temperature: standard 105°C, on request 210°C (Arlon).

If you need a special material (e.g. halogen-free, high-Tg, metal-core, BT epoxy or PTFE), please contact us. You can find an overview on our website: www.multi-circuit-boards.eu

Text data contained in this data sheet represents typical values and does not constitute any warranty or guarantee. For review of critical specification tolerances, please contact our CAM department.

SUPPORT:
Multi Leiterplatten GmbH
Brunnthaler Straße 2
DE-85649 Brunnthal
+49 8104 628-0 | Fax: -160

CONTRACTING PARTY:
Multi Circuit Boards Ltd. | Business Centres, Holyrood Close | GB-Poole, BH17 7FJ
VAT-ID: DE281397430 | GB Co. Reg: 07909372
Contact: Tel: +44 1425 489 111 | info@multi-circuit-boards.eu | www.multi-cb.eu
Bank details: Commerzbank Munich | IBAN DE31 7004 0048 0763 3407 00 | BIC COBADEFF970

FED Fachverband für Design, Leiterplatten- & Elektronikfertigung
IPC Member